

Text Analytics 635AA

A.Y. 2025/26 Written test July, 17, 2026

Deliver results within 2 h from start time. No teaching material is allowed during the test

Exercise 1 [8 pts].

- (i) Describe the Naïve Bayes classifier for text classification. Explain how it is derived from Bayes' theorem, and identify the roles of the prior probability, the likelihood, and the posterior probability. State clearly the "naïve" assumption, and explain how the prior of each class and the per-word likelihoods are estimated from a labelled training corpus. Finally, describe how a new document is classified. (4 points)
 - (ii) Explain the zero-probability problem that arises when a word never occurs in the training documents of a given class, and why it is especially damaging for Naïve Bayes. Describe how Laplace smoothing addresses it, and explain why the computation is usually carried out in the log domain. (2 points)
 - (iii) Discuss the main strengths and limitations of Naïve Bayes for text. Comment on the realism of the "naïve" assumption, and briefly contrast Naïve Bayes with Logistic Regression. (2 points)
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Exercise 2 [8 pts].

- (i) Describe the structure of the perceptron: how it combines its inputs through weights and a bias term and how the result is passed through an activation function. Explain the purpose of non-linear activation functions and why non-linearity is essential. Then describe how neurons are organized into a multi-layer feedforward neural network, and explain what is meant by the forward pass. (4 points)
 - (ii) Explain how a feedforward neural network is trained. Describe the role of the loss function, the idea of gradient descent as an optimization procedure, and the role of backpropagation in computing the gradients used to update the weights. (2 points)
 - (iii) Explain what is meant by overfitting and underfitting when training a neural network. Mention at least one strategy that can be used to mitigate overfitting. (2 points)
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Exercise 3 [8 pts].

- (i) Describe the Recurrent Neural Network (RNN) architecture for processing sequential text. Explain how the hidden state is computed and propagated across time steps to build a representation of the sequence, and how it differs from a feedforward network that processes each input independently. Explain what is meant by *unfolding* (unrolling) the network over time and how training is carried out. (3 points)
 - (ii) Explain why vanilla RNNs struggle to capture long-range dependencies. Describe the vanishing (and exploding) gradient problem and its consequences for learning from long sequences. (2 points)
 - (iii) Describe the Long Short-Term Memory (LSTM) architecture and how it mitigates the vanishing-gradient problem. (3 points)
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Exercise 4 [8 pts].

- (i) Explain what topic modeling is and why it is an unsupervised task. Describe the progression from representing a topic as a single term to representing a topic as a probability distribution over the whole vocabulary, and explain why the latter is a richer representation. Explain in what sense a document is modeled as a mixture of topics, and briefly contrast topic modeling with document clustering. (4 points)
- (ii) Describe the main intuition behind Latent Dirichlet Allocation (LDA) as a generative probabilistic model. Outline how each document is associated with a distribution over topics and how each word is generated by first choosing a topic and then drawing a word from that topic's word distribution. (2 points)
- (iii) Explain how the output of a fitted LDA model is represented and interpreted: what a topic looks like in practice, and how a document is described once the model has been trained. Describe at least one practical application of topic modeling in a text analytics context. (2 points)